In-Class Template

Alex Gonzales

1. Set Up

Packages

```r
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)


# wrap axis labels
library(scales)


# reading .xlxs files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```r
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")


# site information
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  # powerful argument in read_xlsx to show the missing variables as numbers
  na = c("", "n/a", "N/A", "over", "173+"))
```

2. Cleaning

`sites` object

Goal: create a new object from `sites` that only includes NCOS sites

Use: filter downstream data frames using new object

```r
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

Cleaning `taxon_list`

Goal: convert character strings in the existing `taxon_list` data frame into snakecase

```r
# creating new object from taxon list
taxon_list_clean <- taxon_list |>
  # converted tacon names to snake case
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))
```

Clean `water_quality`

Goal: create a data frame with mean pH, mean DO (mg/L), mean salinity (ppt) for each date and site

- create a new object from `water_quality`
- clean column names
- only include surveys from quarterly NCOS sites
- create a new column called `date_site` to join with later data frames
- group by date_site
- calculate median pH, salinity, and dissolved oxygen
- ungroup data frame
- filter out any outliers

```r
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

Cleaning `aq_ins`

Goal: create long format data frame in which each row contains the counted individuals in each taxon during each survey on each date with all the taxonomic information for the taxon and the water quality information for the survey

Use: visualize relationships between water quality parameters and aquatic invertebrate abundances

Pseudocode:

- create new object from `aq_ins`
- clean column names
- filter to only include sites in NCOS sites
- select columns of interest
- filter to only include "filtered beaker" samples
- convert the data frame to long format
- group by date, site, taxon
- calculate average abundance per liter (sum abundance divided by 7.5 liters)
- create a new column called `date_site` to join with water quality
- join with water quality
- join with taxon list

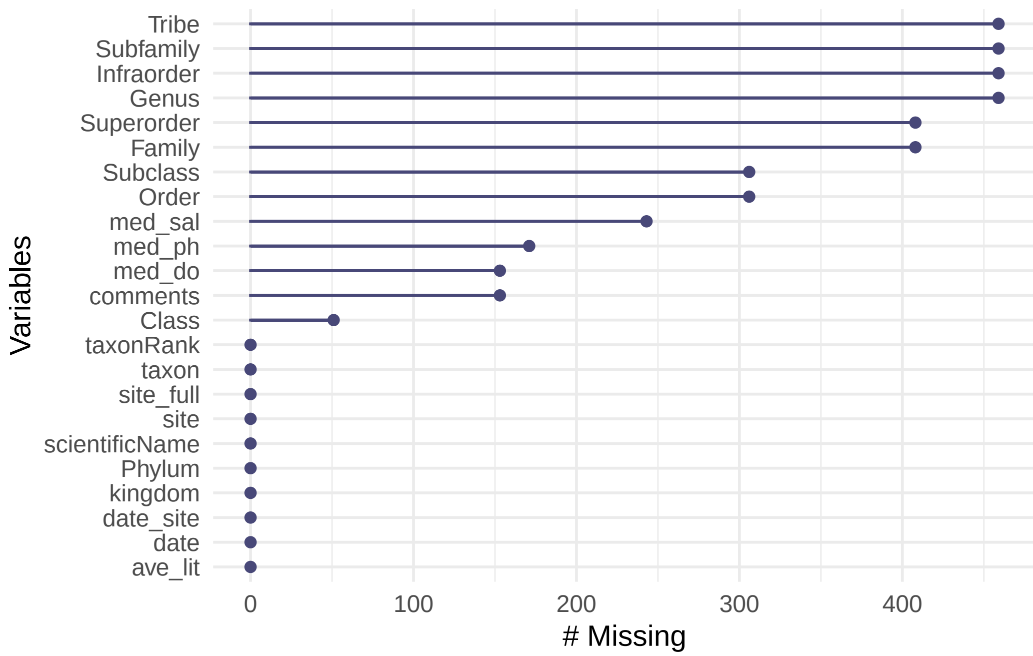
```
# creating new object
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
# filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
```

```
# convert the data frame to long format
  pivot_longer(
    # columns to make longer
    cols = ostracod:annelida_polychaete,
    # new column for the old column names
               names_to = "taxon",
    # new column for the values in each of the old columns
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5 liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # created new column to join with data frames
  unite("date_site", date, site, remove = FALSE) |>
  # join with water_quality_clean
  left_join(water_quality_clean, by = "date_site") |>
  # joined with taxon_list_clean
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # fill in full site names
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "East Channel",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for full site name by median salinity
  mutate(site_full = fct_reorder(site_full,
                                 med_sal,
                                 .fun = "median",
                                 .na_rm = TRUE),
         site_full = fct_rev(site_full))
```

3. Visualizing

Visualizing Missing values

```
gg_miss_var(aq_ins_clean)
```



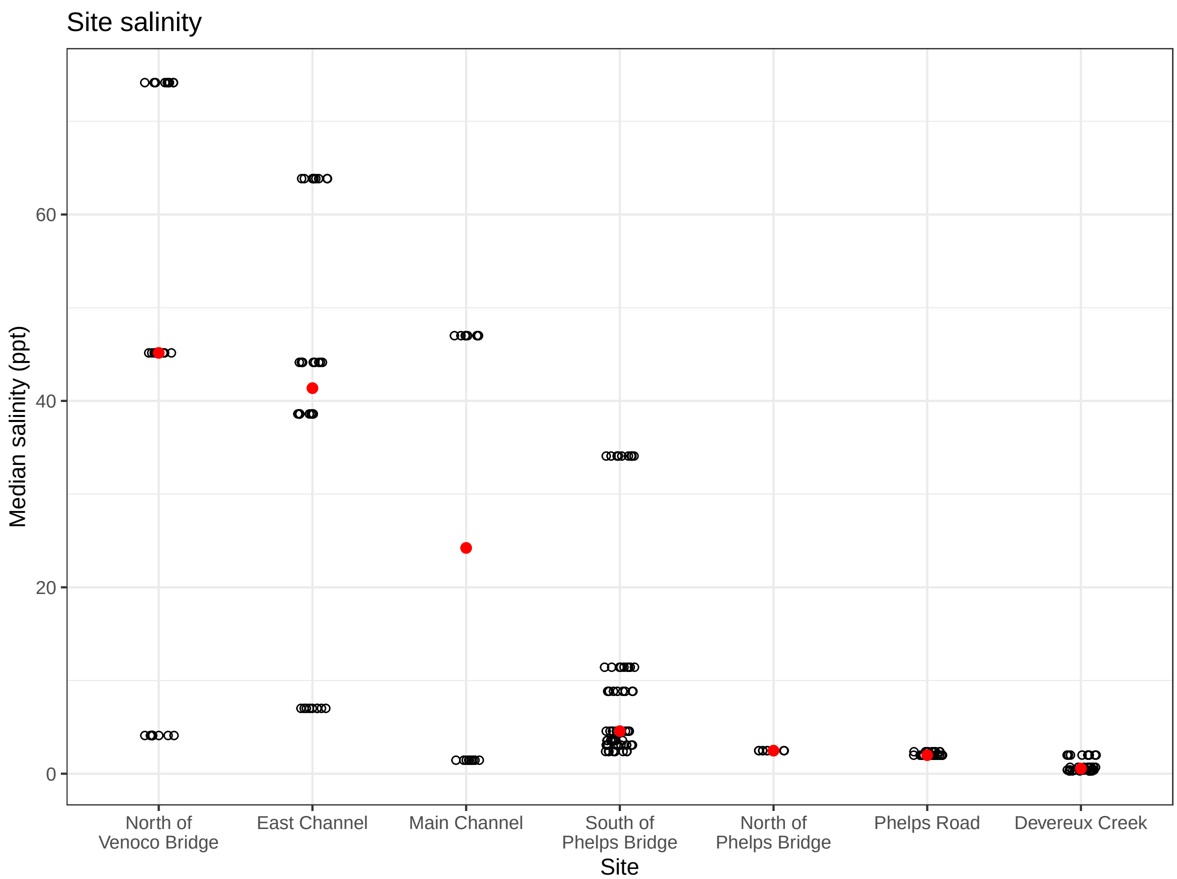
Missing values for salinity, pH, and dissolved oxygen might influence later figures

Visualize salinity

- x-axis: `site`
- y-axis: `mean_sal`
- geometry to show salinity during survey: `geom_jitter()`
- geometry to show mean salinity: `stat_summary()`

```
# base layer
salinity_plot <- ggplot(data = aq_ins_clean,
       # x-axis: full site name
       # y-axis: median salinity
       mapping = aes(x = site_full,
                     y = med_sal)) +
```

```
  # first layer: salinity measurements at each survey
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # points representing median salinity
  stat_summary(geom = "point",
               fun = median,
               color = "red",
               size = 2) +
  #  wrapping x-axis labels
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # relabeling axes, adding title
  labs(x = "Site",
       y = "Median salinity (ppt)",
       title = "Site salinity") +
  # using different theme from default
  theme_bw()

# displaying plot
salinity_plot
```



Sites seem to differ in salinity. North of Venoco Bridge, East Channel, and Main Channel tend to be hypersaline, while North of Phelps Bridge, Phelps Road, and Devereux Creek is the least saline. South of Phelps Bridge tends to be more freshwater but is higher in salinity than the other, lower salinity sites.

Visualize differences in copepod abundance

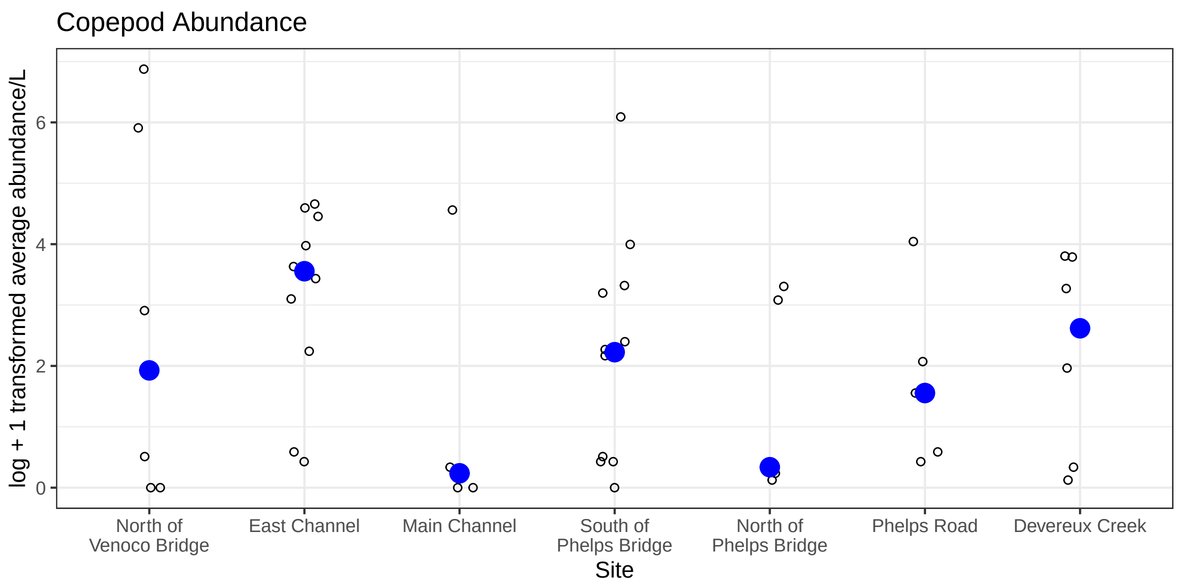
- x-axis: `site_full`
- y-axis: `ave_lit`
- geometry to represent observations: `geom_jitter()`
- geometry to represent medians: `stat_summary()`

Initial Cleaning

```r
copepods <- aq_ins_clean |>
  filter(taxon == "copepod") |>
 # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
```

Visualizing:

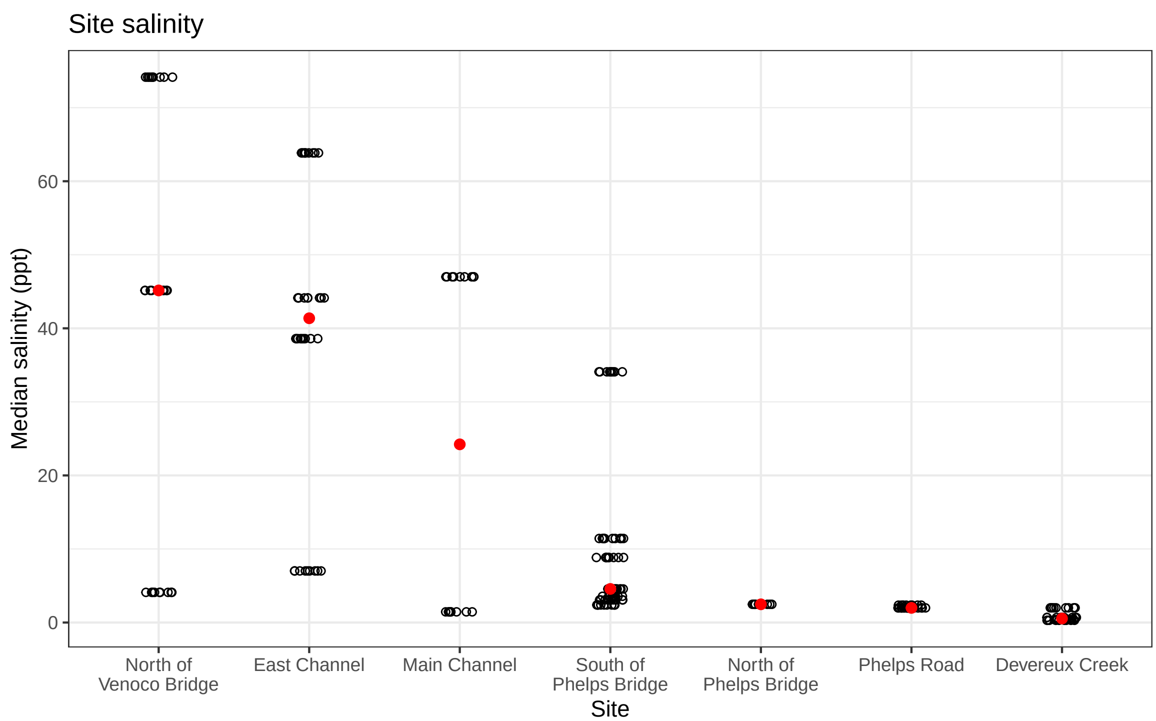
```r
# base layer, creating oobject to store plot
copepod_plot <- ggplot(data = copepods,
                   # x-axis: full site name
                   # y-axis: log transformed average count/L
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
  # first layer: points to represent observations, jitter to represent each sample
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepod Abundance") +
  # using different theme
  theme_bw()

# displaying object
copepod_plot
```

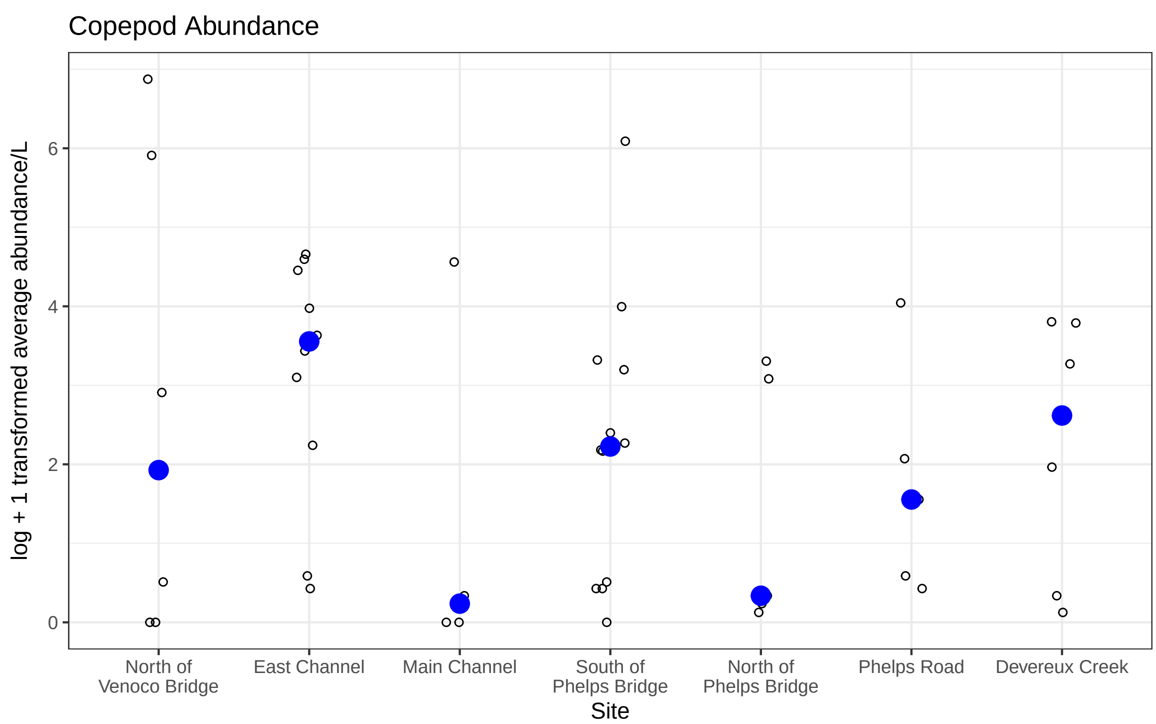


Copepod abundance differs across sites.

Putting salinity and copepod abundance together

```
salinity_plot / copepod_plot
```


Site salinity
Median salinity (ppt)
0
20
40
60
North of
Venoco Bridge
East Channel
Main Channel
South of
Phelps Bridge
North of
Phelps Bridge
Phelps Road
Devereux Creek
Site


Copepod Abundance
log + 1 transformed average abundance/L
0
2
4
6
North of
Venoco Bridge
East Channel
Main Channel
South of
Phelps Bridge
North of
Phelps Bridge
Phelps Road
Devereux Creek
Site